Kai Liu

Academic Activity List

Postdoctoral Scholar
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Upcoming Presentations

1. **Invited talk.** "Frame phase retrievability and state distinguishability of quantum channels." Special Session on New Faces in Operator Theory, *2027 Joint Mathematics Meetings*, Chicago, IL, January 12–15, 2027. Event
2. **Invited talk.** "Phase Retrievability of Quantum Channels and Interferometric Coupling." Special Session on Mathematical Aspects of Quantum Information and Computation, IV, *2026 Fall Western Sectional Meeting*, Arizona State University, Tempe, AZ, November 8, 2026. Program
3. **Invited talk.** "Dynamical Sampling for Pure State Identification in Open Quantum Systems." Special Session on Recent Developments in Applied Harmonic Analysis, II, *2026 Fall Western Sectional Meeting*, Arizona State University, Tempe, AZ, November 7, 2026. Program

Conference and Workshop Presentations

4. **Invited talk.** "Operator valued frames and phase retrieval of quantum channels." *9th International Conference on Computational Harmonic Analysis (ICCHA 2026)*, in conjunction with the 38th Shanks Lecture, Vanderbilt University, Nashville, TN, May 21, 2026. Program
5. **Invited talk.** "Enhance Phase Retrievability via Quantum Interferometer." Special Session on Frames, Sampling, and Quantum Information Theory, II, *2026 Spring Southeastern Sectional Meeting*, Georgia Southern University, Armstrong Campus, Savannah, GA, March 28, 2026. Program
6. **Contributed talk.** "Zero-error Correctibility and Phase Retrievability for Twirling Channels." AMS Contributed Paper Session on Computer Science, Information, and Communication, I, *2026 Joint Mathematics Meetings*, Washington, DC, January 4, 2026. Program
7. **Invited talk.** "Frame Theory in Phase Retrievable POVMs and Quantum Channels." Special Session on Inverse Problems IV, *2025 Fall Central Sectional Meeting*, Saint Louis University, St. Louis, MO, October 19, 2025. Program
8. **Invited talk.** "Phase Retrievability of Frames and Quantum Channels." AMS Special Session on Recent Progress in Frame Theory and Harmonic Analysis, I, *2025 Fall Western Sectional Meeting*, University of Denver, Denver, CO, August 23, 2025. Program
9. **Invited talk.** "Fidelity Preservation for Mixed Unitary Quantum Channels." *Workshop on New Classes of Spectral Duality, Inverse Problems, Operators, Analysis, and Algorithms*, online, March 22–23, 2025.

Colloquia and Seminars

10. **Invited talk.** "Frames and Generalized phase damping quantum channels." October 2024. Colloquium organized by the Department of Mathematics, UCF.
11. **Invited talk.** "Fidelity preserving phenomenon under mixed unitary quantum channels." April 19, 2024. Mathematics and Applications Seminar organized by the Department of Mathematics, UCF. Event